Yue Qiao 乔越

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Audio algorithms in two directions: loudspeaker arrays to reproduce and enhance what people hear, and microphone arrays to sharpen how devices sense space.
Doctoral work on sound field control and binaural rendering. Now building multi-microphone speech front ends for consumer devices, with the plan to bring both sides to human-machine interaction in real environments.

EXPERIENCE

Huawei Technologies 2012 Laboratories — Dec 2024 — Present
Audio Algorithm Engineer · **Huawei Top Minds Program**（天才少年）

- Own multi-microphone speech front-end processing for wearable devices: hearing assistance and conversation enhancement, calls, AI voice interaction, directional recording, and voiceprint-driven speech enhancement.
- Design the capture algorithm chain and the network architecture. Deploy on a single-core edge NPU under tight memory and compute budgets, and trace accuracy loss from quantization, fixed-point conversion, and simulation-to-device gaps.
- Led spatial audio rendering for automotive cabins, from prototype design and validation through delivery. Shipped in the **Maextro S800**.
- Build training data from acoustic simulation to validate prototype capture algorithms across device categories. Guide microphone hardware layout from the results.
- Lead two university collaborations on device-side speech separation: one selects the target talker by identity and distance on a microphone array, the other drives separation with natural language instructions.

Tencent AI Lab — Jun 2024 — Aug 2024
Research Intern, Bellevue, WA

- Neural Ambisonic encoding on a circular microphone array for multi-speaker scenes. Published at ICASSP 2025.

Princeton University, 3D3A Laboratory — 2019 — 2024
Assistant in Research, advisor: Edgar Choueiri

- Head-tracked personal sound zone rendering, combining deep learning and classical DSP.
- Sponsored projects: head-tracked loudspeaker beamforming (Focal), cabin sound stage control (Tesla).

Reality Labs, Meta — May 2022 — Aug 2022
Research Engineer Intern, Sunnyvale, CA

- Machine learning models that predict subjective spatial audio quality, replacing part of the listening tests. Published at AES Convention 153.

SKILLS

Speech front end · Multi-microphone array processing, beamforming (including neural beamforming), acoustic echo cancellation (AEC), voice activity detection (VAD), speech enhancement, speech separation, voiceprint-driven target speaker extraction, direction and distance estimation

Spatial audio and sound field · Spatial audio rendering, Ambisonics encoding and decoding, binaural rendering with head tracking, personal sound zones and crosstalk cancellation, automotive cabin sound field control

On-device engineering · Edge NPU deployment, model quantization and fixed-point conversion, architecture design under compute and memory budgets, streaming low-latency inference, acoustic simulation and training data construction

Tools · Python, PyTorch, C++, MATLAB, COMSOL

SELECTED PUBLICATIONS

Multichannel array processing and neural spatial modeling

Yue Qiao, V. Kothapally, M. Yu, D. Yu. Neural Ambisonic Encoding for Multi-Speaker Scenarios Using a Circular Microphone Array. ICASSP 2025.

Yue Qiao, E. Choueiri. SANN-PSZ: Spatially Adaptive Neural Network for Head-Tracked Personal Sound Zones. IEEE/ACM TASLP, 2025.

M. Zhang, **Yue Qiao**, X. Wu, T. Qu. Distance-Dependent Modeling of Head-Related Transfer Functions. ICASSP 2019.

T. Qu, Z. Huang, **Yue Qiao**, X. Wu. Matching Projection Decoding Method for Ambisonics System. ICASSP 2018.

Metrics, datasets, and sound field control

Yue Qiao, L. Guadagnin, E. Choueiri. Isolation Performance Metrics for Personal Sound Zone Reproduction Systems. JASA Express Letters, 2022.

Yue Qiao, R. M. Gonzales, E. Choueiri. A Multi-Loudspeaker Binaural Room Impulse Response Dataset with High-Resolution Head Coordinates. Frontiers in Signal Processing, 2024.

Yue Qiao, N. Zacharov, P. F. Hoffmann. Prediction of Timbral and Spatial Audio Quality with Independent Auditory Feature Mapping. AES Convention 153, 2022.

Yue Qiao, J. Luo, E. Choueiri. Spatial Sampling of Binaural Room Transfer Functions for Head-Tracked Personal Sound Zones. JAES, 2024.

Dissertation: Personal Sound Zone Rendering with Listener Individualization and Head Tracking, Princeton University, 2024. Full list on Google Scholar.

PATENTS

A System and Method for Rendering Personal Sound Zones with Head Tracking Using Multiple Loudspeakers. PCT application WO 2026090480A1, published. First inventor. Applicant: Princeton University. Priority date Oct 2024.

Three Chinese invention patents under examination, one as first inventor. Areas: speech separation, spatial audio capture, microphone array capture.

EDUCATION

Princeton University 2024
Ph.D. and M.A., Mechanical and Aerospace Engineering · Advisor: Edgar Choueiri

Peking University 2019
B.S., Physics

HONORS

President Award (individual), Huawei 2012 Laboratories 2026

Innovation and Entrepreneurial Mindset Award, Princeton University 2024

AES Educational Foundation Scholarship 2023

Best Student Paper Award, AES 152nd Convention 2022